

Rule-Based Modeling of Cell Signaling

Module 2: Molecular structure and combinatorial complexity

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MSCBIO 2025 · Introduction to Bioinformatics Programming in Python

Fall 2026

Resources

- ▶ **BioNetGen:** software, documentation, and tutorials: bionetgen.org
- ▶ **VS Code extension:** search the Extensions panel for *BioNetGen*
- ▶ **bngsim:** fast ODE and stochastic simulation of BioNetGen models from Python: bngsim.readthedocs.io
Install with `pip install bngsim bionetgen`
- ▶ **Course repository:**
github.com/jrfaeder/RBM_Intro_BioNetGen
 - `models/` BNGL models used in the slides
 - `notebooks/` Jupyter notebooks for simulation and fitting
 - `Slides/pdf/` slides and handouts



Course repository
models, notebooks, slides

Goals for this module

By the end of this module you should be able to

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2. represent molecules with **components**, **states**, and **bonds** in BNGL
3. read and write **patterns**, and use them to define observables
4. write **reaction rules** and predict which reactions a rule generates
5. describe how BioNetGen **generates a network** from rules, and when network-free simulation is needed instead

Part 1

From species to molecules

Where Module 1 left off

Our cytokine–STAT–SOCS model had **12 species**, each named by hand:

L	R	LR	S
LRS	LRS _p	S _p	G
GS _p	mSOCS	SOCS	LR·SOCS

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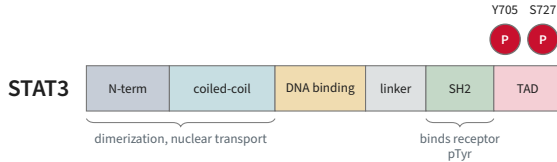
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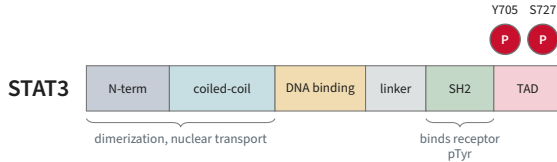
- ▶ Ligand only leaves an empty receptor
- ▶ SOCS and STAT compete for one receptor site
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Each of these is false for real cytokine receptors. Relaxing any of them requires new species, and new copies of every reaction they take part in.

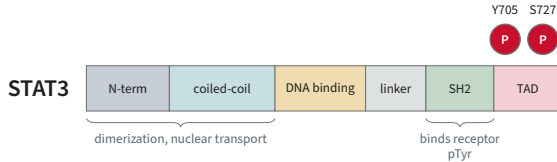
Signaling proteins have many interaction sites



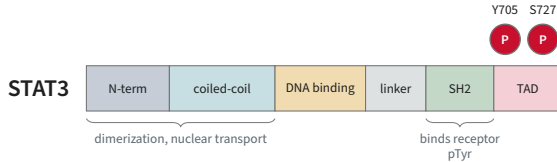
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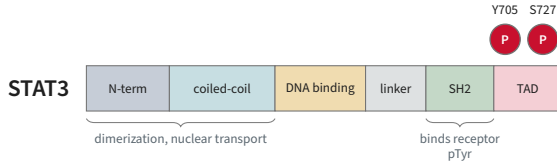
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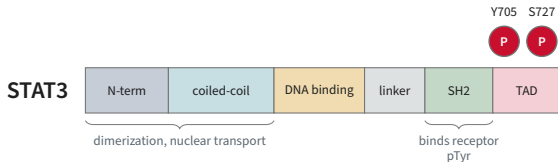
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Signaling proteins have many interaction sites



n independent two-state sites

$\Rightarrow 2^n$ possible states of one molecule

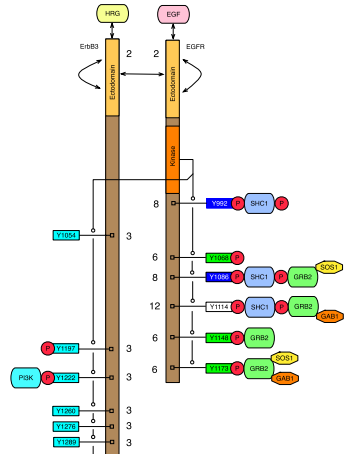
and complexes multiply the counts of their parts

- ▶ Each protein has several **binding domains** and **modification sites**.
- ▶ Sites can be occupied or modified *in combination*.
- ▶ Most interactions depend on only one or two sites: SOCS binding does not care whether S727 is phosphorylated.
- ▶ Naming every combination is **combinatorial complexity**: a few interactions imply a huge number of species.

Combinatorial complexity in a realistic model of EGFR signaling

A model of the ErbB receptor family with its known binding partners:

- ▶ ErbB3:ErbB1 dimer: $> 3.8 \times 10^9$ states

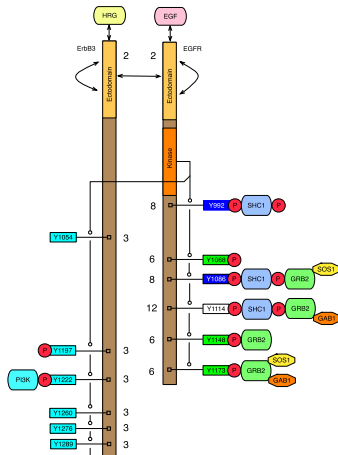


Creamer et al., *BMC Syst. Biol.* 6, 107 (2012)

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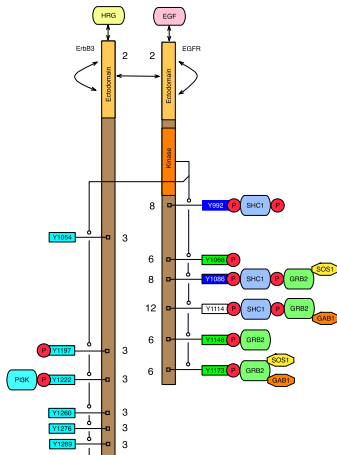
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Yet the model is built from a manageable number of **rules**, in the tens to hundreds.

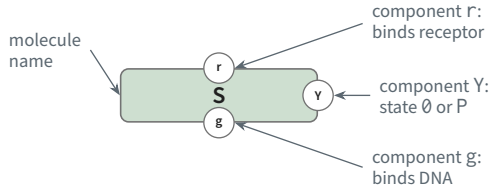


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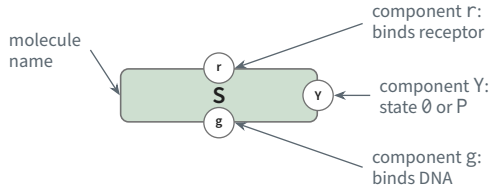
Part 2

Structured molecules in BNGL

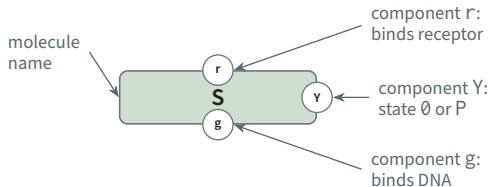
Molecules have components, and components can have states



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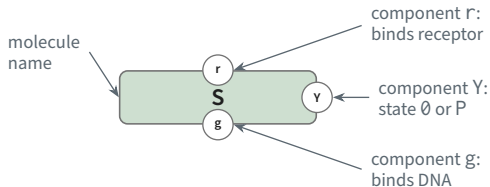
Molecules have components, and components can have states



- ▶ A **molecule** is a named object with a list of **components** in parentheses.
- ▶ Components represent **binding sites** or **modification sites**.
- ▶ $\sim$ lists the allowed **states** of a component: $Y \sim \emptyset \sim P$.

```
begin molecule types
  S(r,Y~ $\emptyset$ ~P,g)
end molecule types
```

Molecules have components, and components can have states



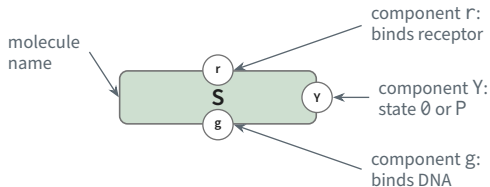
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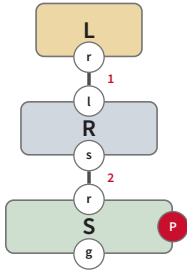
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- ▶ `~` lists the allowed **states** of a component: `Y~ $\emptyset$ ~P`.
- ▶ A component can have both a state and a bond.
- ▶ A molecule with no components, `mSOCS()`, is just a species name, as in Module 1.

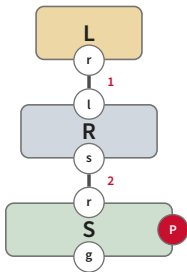
Complexes are molecules joined by bonds



The Module 1 species LRS_p written with structure:

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L(r!1).R(l!1,s!2).S(r!2,Y~P,g)
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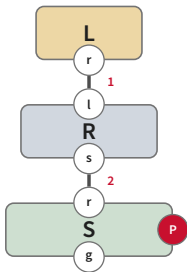


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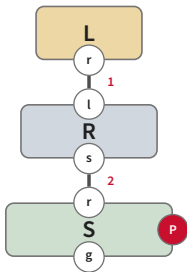


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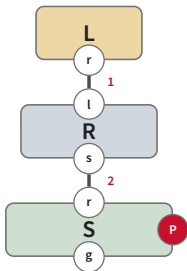


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- ▶ A component with no `!` is **free**: `g` above.
- ▶ A species is now a **graph**: molecules are nodes, bonds are edges. BioNetGen recognizes when two strings describe the same graph.

Patterns select sets of species

A **pattern** lists only the parts of a complex you care about. **Don't write, don't care**: anything not mentioned can be in any state.

pattern	LR	LRS	LRS_p	S_p	GS_p	LR·SOCS	meaning
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S(r!+)	×	✓	✓	×	×	×	STAT bound via r

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S(r!+)	×	✓	✓	×	×	×	STAT bound via r
R(l!+)	✓	✓	✓	×	×	✓	ligand-bound receptor

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R(l!+)	✓	✓	✓	×	×	✓	ligand-bound receptor
R(l!+, s)	✓	×	×	×	×	×	... with docking site free

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R(l!+)	✓	✓	✓	×	×	✓	ligand-bound receptor
R(l!+, s)	✓	×	×	×	×	×	... with docking site free
R(s!+)	×	✓	✓	×	×	✓	docking site occupied

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R(l!+)	✓	✓	✓	×	×	✓	ligand-bound receptor
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- ▶ !+ bound to anything !? bound or not no ! must be free
- ▶ Matching is done on the graph, so the pattern can match a molecule anywhere inside a complex.

Observables are patterns

Module 1: list every species by hand

Molecules pSTAT LRSp Sp GSp

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The pattern keeps working when we add new complexes containing pSTAT. No observable needs to be rewritten.

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Molecules VS. Species

- ▶ Molecules counts every *match*. A STAT dimer with two pTyr contributes 2 to S(Y~P).
- ▶ Species counts each matching *complex* once.

Species pSTATdimer S(d!1).S(d!1)

Rules transform patterns

Sbind: $R(l!+,s) + S(r,Y\sim 0) \leftrightarrow R(l!+,s!1).S(r!1,Y\sim 0) \quad k_{\text{Son}}, k_{\text{Soff}}$

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What a rule can do

form a bond	$A(b) + B(a) \rightarrow A(b!1).B(a!1)$
break a bond	$A(b!1).B(a!1) \rightarrow A(b) + B(a)$
change a state	$S(Y\sim 0) \rightarrow S(Y\sim P)$
make a molecule	$G(s!+) \rightarrow G(s!+) + mSOCS()$
delete one	$SOCS(r) \rightarrow \emptyset$

Part 3

The Module 1 model as rules

Molecule types, seed species, and observables

```
begin molecule types
```

```
  L(r)
```

```
  R(l,s)
```

```
  S(r,Y~0~P,g)
```

```
  G(s)
```

```
  mSOCS()
```

```
  SOCS(r)
```

```
end molecule types
```

```
begin seed species
```

```
  L(r)      L0
```

```
  R(l,s)    R0
```

```
  S(r,Y~0,g) S0
```

```
  G(s)      G0
```

```
end seed species
```

```
begin observables
```

```
  Molecules pSTAT S(Y~P)
```

```
  Molecules SOCS  SOCS()
```

```
end observables
```

Molecule types, seed species, and observables

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  L(r)
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  SOCS(r)
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begin seed species
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  L(r)      L0
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end seed species
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```
begin observables
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end observables
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- Six molecule types replace twelve species names.

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```
  mSOCS()
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```
  SOCS(r)
```

```
end molecule types
```

```
begin seed species
```

```
  L(r)      L $\emptyset$ 
```

```
  R(l,s)    R $\emptyset$ 
```

```
  S(r,Y~ $\emptyset$ ,g) S $\emptyset$ 
```

```
  G(s)      G $\emptyset$ 
```

```
end seed species
```

```
begin observables
```

```
  Molecules pSTAT  S(Y~P)
```

```
  Molecules SOCS   SOCS()
```

```
end observables
```

- ▶ Six molecule types replace twelve species names.
- ▶ **Seed species** are the complexes present at $t = 0$. Every other species is generated from them by the rules.

Molecule types, seed species, and observables

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begin molecule types
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  R(l,s)
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```
  mSOCS()
```

```
  SOCS(r)
```

```
end molecule types
```

```
begin seed species
```

```
  L(r)          L0
```

```
  R(l,s)        R0
```

```
  S(r,Y~0,g)    S0
```

```
  G(s)          G0
```

```
end seed species
```

```
begin observables
```

```
  Molecules pSTAT S(Y~P)
```

```
  Molecules SOCS  SOCS()
```

```
end observables
```

- ▶ Six molecule types replace twelve species names.
- ▶ **Seed species** are the complexes present at $t = 0$. Every other species is generated from them by the rules.
- ▶ The receptor has one docking site s that STAT and SOCS compete for. This is the Module 1 assumption.

File: models/stat_rbm1_rules.bngl

The rules

LRbind:	$L(r) + R(l,s) \leftrightarrow L(r!1).R(l!1,s)$	kon, koff
Sbind:	$R(l!+,s) + S(r,Y\sim 0) \leftrightarrow R(l!+,s!1).S(r!1,Y\sim 0)$	kSon, kSoff
Sact:	$R(s!1).S(r!1,Y\sim 0) \rightarrow R(s!1).S(r!1,Y\sim P)$	kact
Srelease:	$R(s!1).S(r!1,Y\sim P) \rightarrow R(s) + S(r,Y\sim P)$	kdiss
Sdephos:	$S(r,g,Y\sim P) \rightarrow S(r,g,Y\sim 0)$	kdephos
Gbind:	$S(r,Y\sim P,g) + G(s) \leftrightarrow S(r,Y\sim P,g!1).G(s!1)$	kGon, kGoff
Transcribe:	$G(s!+) \rightarrow G(s!+) + mSOCS()$	ktx
mRNAdeg:	$mSOCS() \rightarrow \emptyset$	kmdeg
Translate:	$mSOCS() \rightarrow mSOCS() + SOCS(r)$	ktl
SOCSdeg:	$SOCS(r) \rightarrow \emptyset$	kSOCSdeg
SOCSbind:	$R(l!+,s) + SOCS(r) \leftrightarrow R(l!+,s!1).SOCS(r!1)$	kSOCSon, kSOCSoff

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Sbind:	$R(l!+,s) + S(r,Y\sim 0) \leftrightarrow R(l!+,s!1).S(r!1,Y\sim 0)$	kSon, kSoff
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SOCSdeg:	$SOCS(r) \rightarrow \emptyset$	kSOCSdeg
SOCSbind:	$R(l!+,s) + SOCS(r) \leftrightarrow R(l!+,s!1).SOCS(r!1)$	kSOCSon, kSOCSoff

Context encodes the Module 1 assumptions. $R(l,s)$ in LRbind: ligand binds and leaves only when the docking site is empty. $R(l!+,s)$ in Sbind and SOCSbind: only ligand-bound receptors recruit.

Network generation: rules applied to seed species

1. Start with the seed species.

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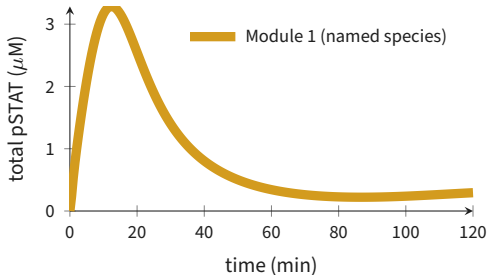
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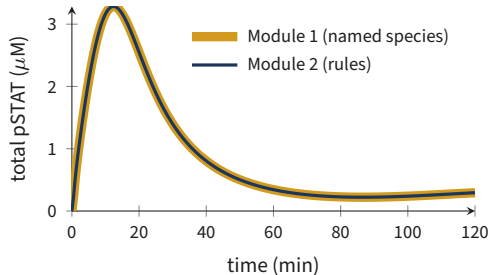


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Result: the same 12 species and 15 reactions as Module 1.



The curves are identical.

Part 4

Relaxing the assumptions

1. Ligand can leave any receptor complex

Remove the context s from the ligand rule:

```
LRbind: L(r) + R(l) <-> L(r!1).R(l!1) kon, koff
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Docking and activation still need bound ligand, but undocking does not, so each reversible rule is split in two:

```
Sbind: R(l!+,s) + S(r,Y~0) -> R(l!+,s!1).S(r!1,Y~0) kSon  
Sunbind: R(s!1).S(r!1,Y~0) -> R(s) + S(r,Y~0) kSoff  
Sact: R(l!+,s!1).S(r!1,Y~0) -> R(l!+,s!1).S(r!1,Y~P) kact
```

(and likewise for SOCS)

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Sact: $R(l!+,s!1).S(r!1,Y\sim 0) \rightarrow R(l!+,s!1).S(r!1,Y\sim P)$ k_{act}

(and likewise for SOCS)

One rule, many reactions

LRbind now generates 4 binding reactions: ligand binds R alone, $R\cdot S$, $R\cdot S_p$, or $R\cdot SOCS$.

	species	reactions
before	12	15

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	species	reactions
before	12	15
after	15	24

2. SOCS binds its own site and blocks kinase activity

Give the receptor a separate SOCS site. SOCS no longer blocks docking; it blocks *activation*:

$R(l,s,socs)$

Sact: $R(l!+,socs,s!1).S(r!1,Y\sim\emptyset) \rightarrow$

$R(l!+,socs,s!1).S(r!1,Y\sim P) \quad k_{act}$

SOCSbind: $R(l!+,socs) + SOCS(r) \rightarrow$

$R(l!+,socs!1).SOCS(r!1) \quad k_{SOCSon}$

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Closer to how SOCS proteins work:
SOCS1 inhibits JAK directly, and SOCS3 inhibits JAK while bound to the receptor.

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	species	reactions
before	15	24
after	19	39

3. Cytokines bring two receptor chains together

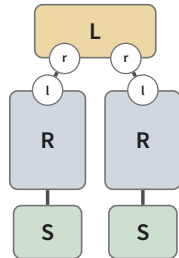
A bivalent ligand $L(r,r)$ crosslinks receptor chains. JAKs on the two chains activate each other, so STAT is activated only in a dimer:

$L(r,r)$

LRbind: $L(r,r) + R(l) \rightleftharpoons L(r!1,r).R(l!1)$ kon, koff

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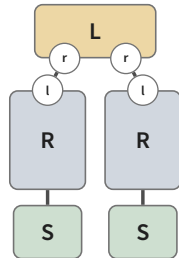
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- BioNetGen accounts for the two identical ligand sites automatically (statistical factor of 2 in LRbind).



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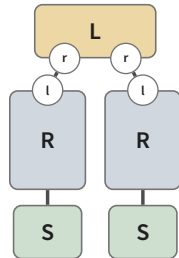
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- ▶ Excess bivalent ligand occupies chains one at a time and blocks crosslinking, so we lower L_0 from 10 to $1 \mu\text{M}$.



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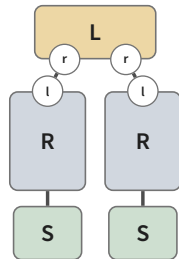
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	species	reactions
before	19	39

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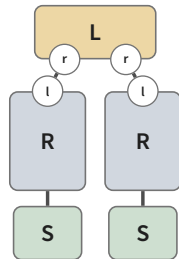
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	species	reactions
before	19	39
after	40	188

Crosslink alone generates 36 reactions in each direction.

4. STATs act as dimers

Add a dimerization site. Free pSTAT dimerizes, only dimers bind the SOCS promoter, and only monomers are dephosphorylated:

$S(r, Y \sim \emptyset \sim P, d, g)$

Sdimer: $S(r, Y \sim P, d) + S(r, Y \sim P, d) \leftrightarrow$

$S(r, Y \sim P, d!1).S(r, Y \sim P, d!1)$ k_{dim}, k_{undim}

Gbind: $S(d!+, g) + G(s) \leftrightarrow S(d!+, g!1).G(s!1)$ k_{Gon}, k_{Goff}

Sdephos: $S(r, d, g, Y \sim P) \rightarrow S(r, d, g, Y \sim \emptyset)$ k_{dephos}

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	species	reactions
before	40	188
after	43	198

5. A second phosphorylation site doubles everything

STAT3 is also phosphorylated on Ser727, independently of Tyr705. One new component and *one* rule:

```
S(r,Y~0~P,S727~0~P,d,g)
```

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Sser: S(S727~0) <-> S(S727~P) kser, kserdephos
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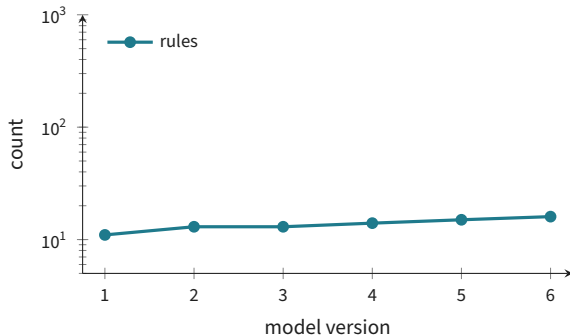
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	species	reactions
before	43	198
after	95	663

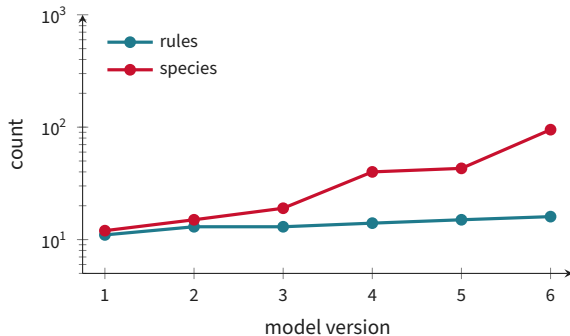
Sser alone generates 59 forward and 59 reverse reactions.

Rules grow linearly; the network grows combinatorially



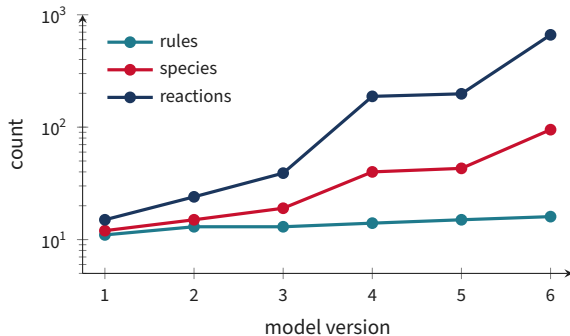
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- 2 ligand leaves any complex
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- 4 receptor dimers
- 5 STAT dimers
- 6 Ser727 site

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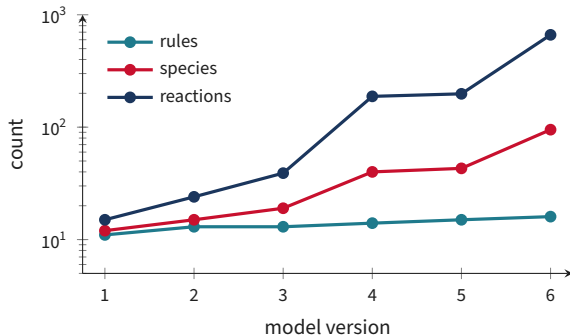
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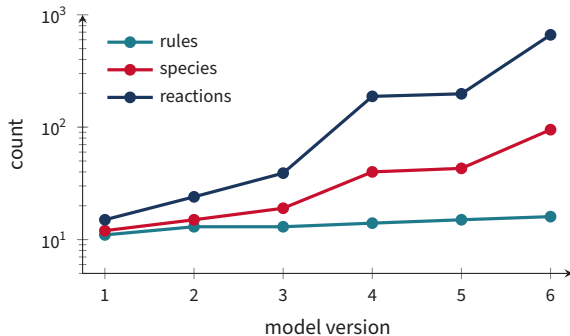
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From 11 to 16 rules, the network grows from 15 to 663 reactions.

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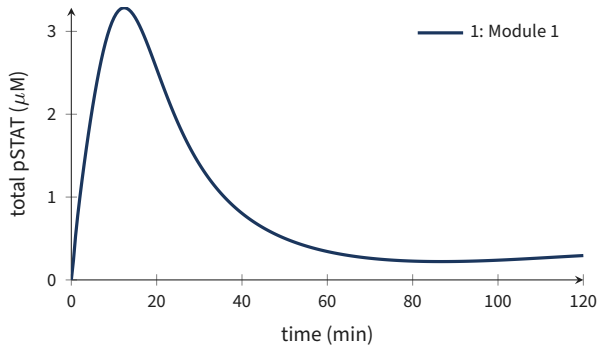


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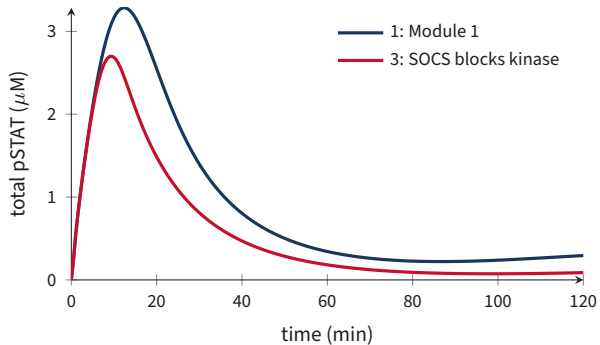
From 11 to 16 rules, the network grows from 15 to 663 reactions.

Writing these 663 reactions, and the ODEs for 95 species, by hand would be slow and error-prone. **Rules let us write down the mechanism and let the software do the bookkeeping.**

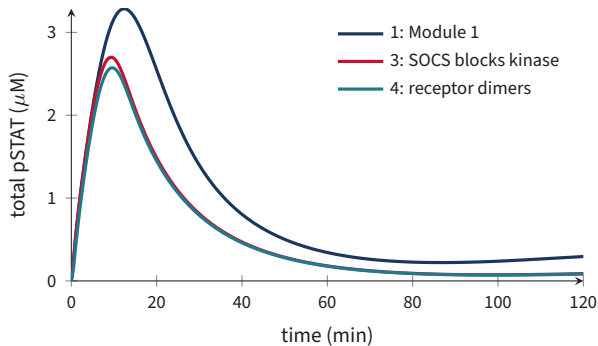
Mechanistic details change the predictions



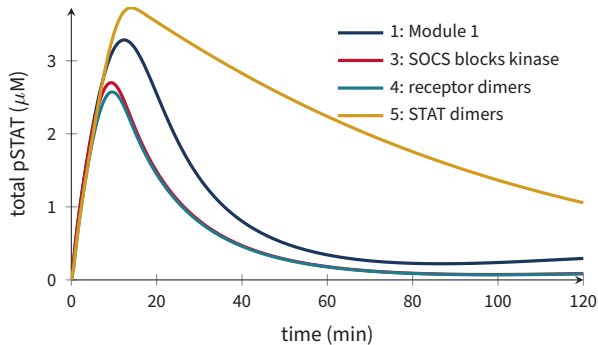
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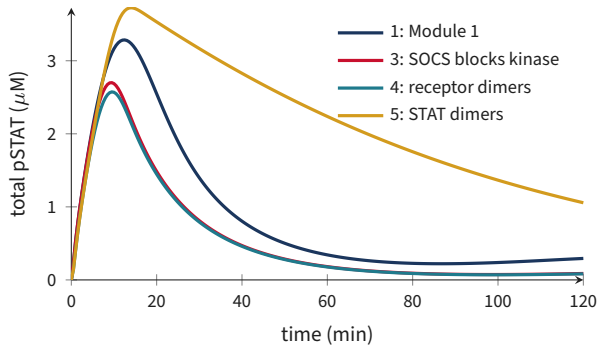
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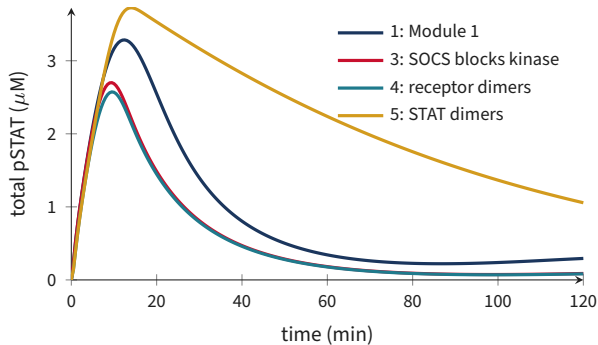


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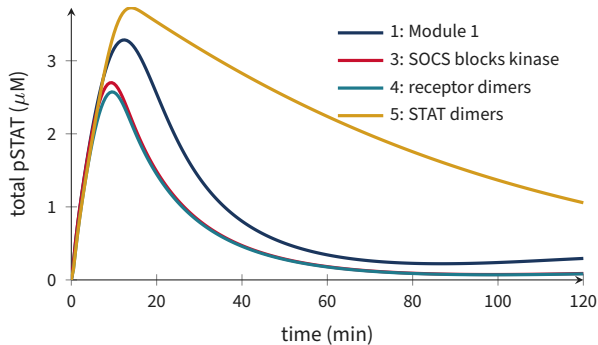
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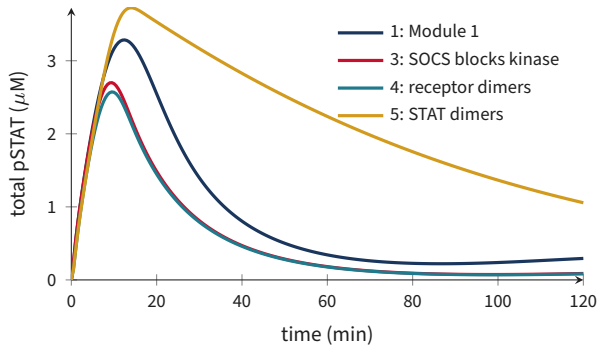
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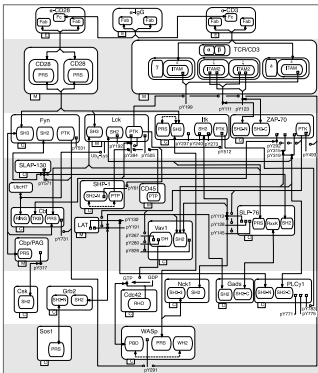
- ▶ All versions show the SOCS-driven pulse.
- ▶ Blocking the kinase (3) gives a lower peak and faster shutoff than blocking docking (1).
- ▶ Dimerization (5) protects pSTAT from the phosphatase, so the response lasts longer.
- ▶ Which mechanism is right is a question for experiments. Rules make it easy to build and compare the alternatives.

Part 5

Scaling up

Rules capture knowledge at large scale

T cell receptor model



**20 molecule types
129 rules**

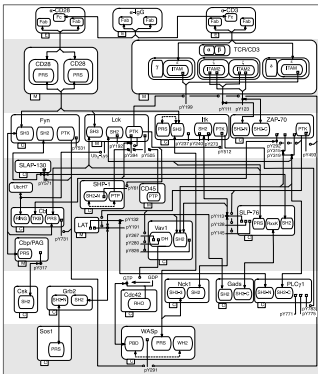
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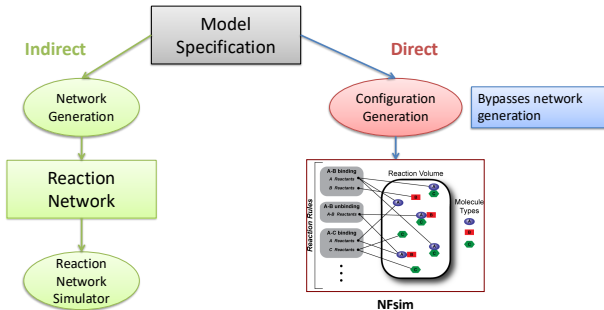
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Rule-based models also serve as precise, extensible, visualizable representations of what is known about a pathway.

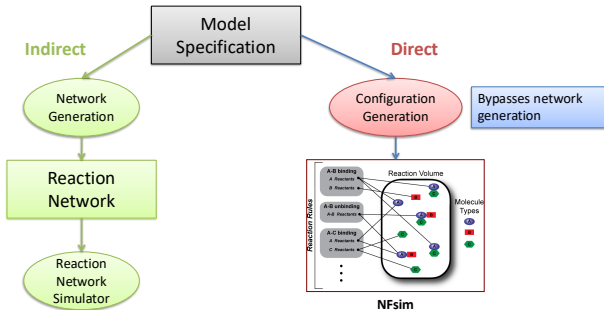
Chylek et al., *PLOS One* (2014)

Two ways to simulate a rule-based model



Indirect: generate the network, then integrate ODEs or run SSA on it. Fast when the network is small. This is what we have done so far.

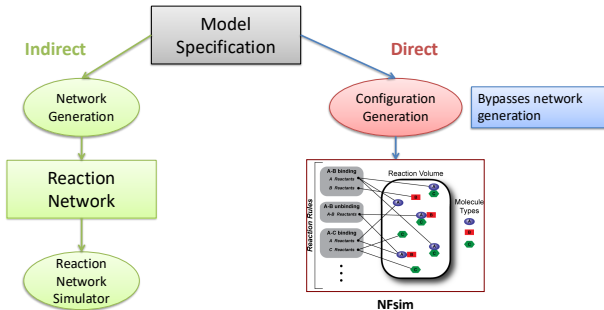
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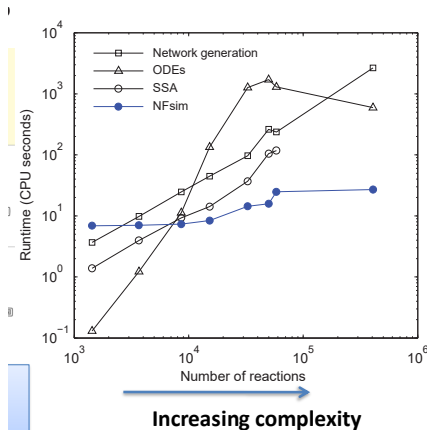


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`simulate({method=>"nf", ...})` runs NFsim.

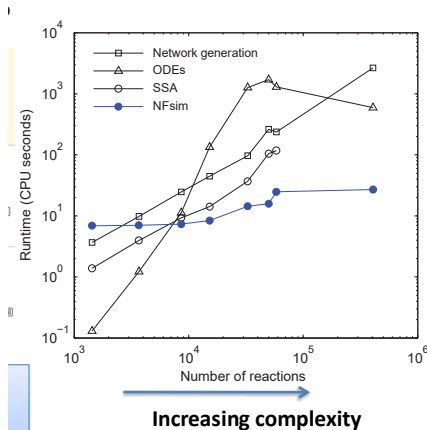
Network-free simulation scales with the rules, not the network



- The cost of network generation and ODE integration grows with the number of reactions.

Sneddon, Faeder, and Emonet, *Nat. Methods* 8, 177 (2011)

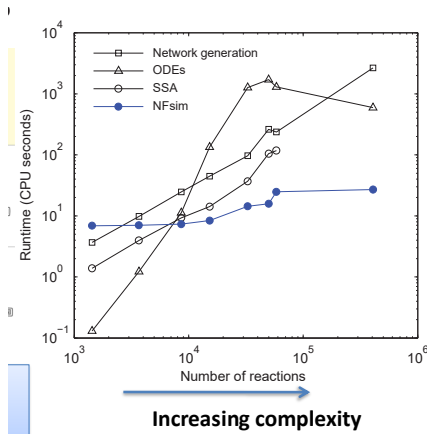
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- ▶ The cost of network generation and ODE integration grows with the number of reactions.
- ▶ NFsim's cost per event depends on the number of rules and molecules, so it handles models whose networks are huge or infinite.
- ▶ Trade-off: NFsim is stochastic and works with molecule counts, so results are noisy and concentrations must be converted to numbers.

Sneddon, Faeder, and Emonet, *Nat. Methods* 8, 177 (2011)

Try it yourself

Using `models/stat_rbm*.bngl`:

1. In `stat_rbm1_rules.bngl`, change `R(1,s)` to `R(1)` in `LRbind`. Generate the network. Which new species appear, and why?
2. Write an observable for receptors that carry both STAT and SOCS in `stat_rbm3_socs_site.bngl`.
3. In `stat_rbm4_receptor_dimer.bngl`, vary `L0` from 0.01 to 100. Plot peak pSTAT against `L0`. Explain the shape.
4. Remove the `r` context from `Sdimer` in `stat_rbm5_stat_dimer.bngl` so receptor-bound pSTAT can dimerize. What happens when you generate the network? Try `generate_network({max_agg=>6})`.

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- ▶ BioNetGen **generates the network** from rules for ODE/SSA simulation; **NFsim** simulates models whose networks are too large to generate.